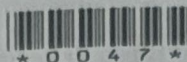


B.Tech. First Year (Second Semester) Mechanical Engineering
Engineering Chemistry
(ME24202)

Total Pages : 3

Time : Three Hours



Max. Marks : 60

Instructions to Candidates :

1. All question carry equal marks.
2. Assume suitable data wherever necessary.
3. Diagrams and chemicals reactions should be given wherever necessary.

A) Solve the following multiple choice questions.

- i) Temporary hardness of water is caused by the presence of 1
 - a) Chlorides of calcium and magnesium
 - b) Sulfates of calcium and magnesium
 - c) Bicarbonates of calcium and magnesium
 - d) Carbonates of calcium and magnesium.
- ii) Estimation of hardness of water by EDTA method is used to determine 1
 - a) Alkaline hardness
 - b) Temporary hardness only
 - c) Permanent hardness only
 - d) All the above

B) Solve **any two** questions of the following.

- i) What is water softening? Explain the process of Zeolite for water Softening. 4
- ii) Explain the EDTA method for determination of hardness of water. 4
- iii) A sample of water is found to contains following dissolving salts in milligrams per litre $\text{Mg}(\text{HCO}_3)_2 = 16.8$, $\text{MgCl}_2 = 12.0$, $\text{MgSO}_4 = 29.6$ and $\text{NaCl} = 5.0$ 4
Calculate temporary and permanent hardness of water

2. A) Solve the following multiple choice questions.

- i) Which of the following is an example of wet corrosion? 1
 - a) Corrosion of metal in the water
 - b) Corrosion of iron in the presence of anhydrous calcium chloride
 - c) Corrosion of titanium in dry chlorine
 - d) Corrosion due to furnace gases
- ii) Which of the following is not a disadvantage of a lithium-ion battery? 1
 - a) Complex to manufacture
 - b) Sophisticated chargers
 - c) More expensive
 - d) High energy density

B) Solve **any two** questions of the following.

- i) Explain construction working and application of lithium ion battery. 4
- ii) Discuss the wet corrosion mechanism by H_2 gas evolution. 4
- iii) What is corrosion? Discuss different types of corrosion. 4

3. A) Solve the following multiple choice questions.

- i) Which of the following is a thermosetting polymer? 1
 - a) polystyrene
 - b) polyolefins
 - c) nylons
 - d) phenolic resins
- ii) Out of the following polymers, which one does not involve cross-linkages? 1
 - a) Vulcanized Rubber
 - b) Polythene
 - c) Melamine
 - d) Bakelite

B) Solve **any two** questions of the following.

- i) Explain the mechanism of cationic polymerization with suitable example. 4
- ii) What are thermosetting polymers? Give preparation, properties and uses of Bakelite. 4
- iii) Explain classification of polymer on the basis of structure and synthesis. 4

4. A) Solve the following multiple choice questions.

- i) Which of the following is a fuel that is good for the environment and is used in automobiles? 1
 - a) Diesel
 - b) Petrol
 - c) Natural gas
 - d) Petroleum gas
- ii) Which are the main constituents of fuel from given options? 1
 - a) Carbon and Nitrogen
 - b) Oxygen and Hydrogen
 - c) Carbon and Hydrogen
 - d) Helium and Oxygen

B) Solve **any two** questions of the following.

- i) What is proximate analysis? Discuss the determination of constituent present on it. 4
- ii) What is the relation between knocking and chemical structure of constituents in gasoline. 4
- iii) What is calorific value of fuel? Explain Gross and Net calorific value of fuel? 4

A) Solve the following multiple choice questions.

- i) The Fe-C phase diagram is an example of a: 1
 a) Unary phase diagram b) Binary phase diagram
 c) Ternary phase diagram d) Quaternary phase diagram
- ii) The eutectic reaction in the Fe-C diagram occurs at: 1
 a) 723°C b) 1147°C
 c) 1493°C d) 1833°C

B) Solve **any two** questions of the following.

- i) Draw a neat labeled diagram of the Fe-C equilibrium diagram and explain its major phases. 4
- ii) Explain annealing and its different types. Discuss its advantages and disadvantages. 4
- iii) What is mechanical metallurgy? Discuss its importance in material behavior and industrial applications. 4

6. A) Solve the following multiple choice questions.

- i) That is the primary purpose of alloying elements in steel? 1
 a) Reduce hardness
 b) Improve mechanical and chemical properties
 c) Increase electrical conductivity
 d) Decrease density
- ii) What is the main alloying element in brass? 1
 a) Tin b) Copper
 c) Zinc d) Aluminium

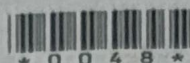
B) Solve **any two** questions of the following.

- i) What is OHNS (Oil Hardening Non-Shrinking) steel? Discuss its composition, properties and applications. 4
- ii) Differentiate between brass and bronze. Compare their composition, properties and applications. 4
- iii) Classify alloy steels based on composition. Describe the key characteristics of each type. 4

B.Tech. First Year (Second Semester) Civil Engineering
Modern Chemistry
(CE24202)

Total Pages : 3

Time : Three Hours



Max. Marks : 60

Instructions to Candidates :

1. Assume suitable data and diagram whenever necessary.
2. You can use scientific calculator during exam.

- A) Solve the following multiple choice questions.
- i) ----- processes is used for the removal of permanent hardness of water. 1
 - a) Boiling
 - b) Clark's process
 - c) Ion-exchange method
 - d) Filtration
 - ii) ----- is the main purpose of adding alum in water treatment? 1
 - a) To kill bacteria
 - b) To remove hardness
 - c) To aid coagulation and sedimentation
 - d) To increase pH
- B) Solve **any two** questions of the following.
- i) Explain Zeolite process for water softening. 4
 - ii) Define Hardness of water? Explain Scale and Sludge. 4
 - iii) Describe Ion Exchange process for water softening. 4
2. A) Solve the following multiple choice questions.
- i) ----- occurs due to the formation of anodic and cathodic regions on the same metal surface? 1
 - a) Galvanic corrosion
 - b) Pitting corrosion
 - c) Crevice corrosion
 - d) Differential aeration corrosion
 - ii) ----- is an example of a sacrificial anode used in cathodic protection? 1
 - a) Copper
 - b) Zinc
 - c) Silver
 - d) Iron
- B) Solve **any two** questions of the following.
- i) Define corrosion? Describe: 4
 - a) Pitting corrosion
 - b) Waterline corrosion
 - ii) Describe construction & working of Li-ion battery. 4
 - iii) Define hot dipping? Explain Galvanizing & Tinning. 4

3. A) Solve the following multiple choice questions.

- i) _____ is an example of a thermosetting polymer?
 a) Polyethylene b) Polyvinyl chloride (PVC)
 c) Bakelite d) Polypropylene
- ii) _____ is commonly used in the manufacturing of plastic bottles?
 a) Polystyrene b) Polyethylene terephthalate (PET)
 c) Polyvinyl chloride (PVC) d) Teflon

B) Solve **any two** questions of the following.

- i) What is Polymer Explain Mechanism of Addition Polymerization.
- ii) Differentiate thermosetting & Thermoplastic.
- iii) Explain Preparation Properties and Use of Bakelite.

4. A) Solve the following multiple choice questions.

- i) _____ compound in cement is responsible for its initial setting?
 a) Tricalcium silicate (C2S) b) Dicalcium silicate (C3S)
 c) Tricalcium aluminate (C3A) d) Tetracalciumaluminoferrite (C4AF)
- ii) The process of heating limestone and clay in a rotary kiln to form cement clinker is called:
 a) Hydration b) Calcination
 c) Polymerization d) Sintering

B) Solve **any two** questions of the following.

- i) Name the Ingredients of cement? Explain their functions.
- ii) Explain Heat of Hydration & Soundness of Cement.
- iii) Describe Wet process of Manufacturing of Cement.

5. A) Solve the following multiple choice questions.

- i) _____ is NOT a function of a lubricant?
 a) Reducing friction b) Preventing wear and tear
 c) Increasing friction d) Cooling moving parts
- ii) _____ lubricant is commonly used in high-temperature applications?
 a) Mineral oil b) Synthetic oil
 c) Water-based lubricant d) Animal fat

B) Solve **any two** questions of the following.

- i) Describe: i) Viscosity ii) Flash point
- ii) Define Lubricant. Explain their functions.
- iii) Explain Thick and Thin Film of Lubricants.

A) Solve the following multiple choice questions.

i) ----- properties is unique to nanomaterials compared to bulk materials? 1

- a) Decreased surface area
- b) Lower reactivity
- c) Increased surface-to-volume ratio
- d) Reduced optical activity

ii) Gibbs Phase Rule is given by the equation? 1

- a) $P + C = F + 2$
- b) $F = C - P + 2$
- c) $P = C - F + 2$
- d) $F = P - C + 2$

B) Solve **any two** questions of the following.

i) State phase rule? Describe application of phase rule to one component system. 4

ii) Describe Fullerenes-properties and uses. 4

iii) Explain synthesis and industrial applications of CNTS. 4

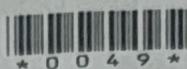
B.Tech. First Year (Second Semester) Artificial Intelligence & Data Science / Computer Science & Engineering / Machine Learning / Mechanical Engineering / Civil Engineering / Electrical Engineering

Computer Programming

(CE24203 / EE24203 / ME24203 / AI24103 / CS24103 / ML24103 / ET24104)

Total Pages : 3

Time : Three Hours



Max. Marks : 60

Instructions to Candidates :

1. All question carry marks as indicated.
2. Assume suitable data wherever necessary.

1. A) Solve the following multiple choice questions.
- i) Which of the following is a fundamental data type in C used to store whole numbers?
a) string b) float
c) int d) boolean
- ii) What is the term for a step-by-step procedure to solve a problem?
a) Program b) Algorithm
c) Code d) Syntax
- B) Solve **any two** questions of the following.
- i) List three fundamental data types commonly used in C programming.
- ii) Discuss Logical & Bitwise operators in C with example.
- iii) Define Algorithm? Explain its characteristics with example.
2. A) Solve the following multiple choice questions.
- i) Which keyword is used to terminate a loop prematurely in C?
a) skip b) end
c) break d) exit
- ii) Which loop in C guarantees that the loop body will execute at least once?
a) for loop b) while loop
c) do-while loop d) if statement
- B) Solve **any two** questions of the following.
- i) List & explain the three types of loop structures commonly used in C programming.
- ii) Write a C program to read the day number & display the day names as an output.

- iii) State the purpose of the 'break' statement with in a loop or switch statement in C. 4
3. A) Solve the following multiple choice questions. 1
- i) What is a collection of elements of the same data type stored in contiguous memory location called? 1
- a) Class b) Object
c) Array d) Function
- ii) What is the output of the following code? 1
- ```
int arr[2][3] = {{1, 2, 3}, {4, 5, 6}};
printf("%d", arr[1][1]);
```
- a) 1 b) 2  
c) 3 d) 5
- B) Solve any two questions of the following.
- i) Write a C program for the subtraction of two matrices. 4
- ii) Define Strings. Explain the different ways of declaring & initializing strings. 4
- iii) Demonstrate array & its types with example. 4
4. A) Solve the following multiple choice questions. 1
- i) What is a self-calling function known as? 1
- a) Iterative function b) Library function  
c) Recursive function d) Main function
- ii) What is the purpose of the void keyword in a function declaration? 1
- a) To indicate that the function returns no value  
b) To indicate that the function takes no arguments.  
c) To indicate that the function is recursive  
d) To indicate that the function is static.
- B) Solve any two questions of the following.
- i) Classify different math functions with explanation & example. 4
- ii) Explain User defined functions? Discuss the four different ways to write user defined functions. 4
- iii) Discuss storage classes with example. 4
5. A) Solve the following multiple choice questions. 1
- i) Which function calling mechanism passes the actual memory address of a variable to the function? 1
- a) Call by value b) Call by reference  
c) Call by function d) Call by array

- ii) What is the output of the following code? 1
- ```
int x = 10;
int *ptr = &x;
printf("%d", *ptr);
```
- a) The address of x b) 10
c) The memory address of ptr d) 10A2F

B) Solve any two questions of the following.

- i) Discuss pointers with example. 4
- ii) Write a C program to demonstrate array of pointers. 4
- iii) Define pointer arithmetic & explain it with example. 4

A) Solve the following multiple choice questions.

- i) What is a user-defined data type that can hold elements of different data types called? 1
- a) Array b) Structure
c) Pointer d) Function
- ii) What is the size of a structure? 1
- a) The sum of the sizes of its members
b) The size of the largest member
c) The size of the first member.
d) The size of the last member.

B) Solve any two questions of the following.

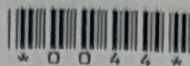
- i) Demonstrate Structures with example. 4
- ii) Explain different File operations in detail with example. 4
- iii) Write a C program to create a structure employee and store the id, name & salary of two employees and display the information. 4

B.Tech. First Year (Second Semester) Electrical Engineering / Civil Engineering /
Mechanical Engineering / Electronics & Telecommunication Engineering

Numerical Methods and Calculus
(EE24101 / CE24101 / ME24101 / ET24101)

Total Pages : 3

Time : Three Hours



Max. Marks : 60

Instructions to Candidates :

1. Illustrate your answer with neat and clear sketch.
2. Assume Suitable data whenever necessary.
3. Use of scientific calculator is allowed.

1. A) Solve the following multiple choice questions.
 - i) What is the expected value (mean) for the Binomial Distribution $n = 16, p = 0.85$ 1
 - a) 7 b) 13.6
 - c) 12.4 d) 6
 - ii) If the mean and variance of Binomial distribution are 6 and 3 respectively then n is 1
 - a) 12 b) 10
 - c) 15 d) 20
- B) Solve **any two** questions of the following.
 - i) If on an average one ship in every 10 is worked, find the probability that out of 5 ships expected to arrive safely, 4 at least will arrive safely. 4
 - ii) If variance of a poison distribution is 2, find the probabilities for $r = 1, 2, 3, 4$ from the recurrence relation of the Poisson distribution. 4
 - iii) A sample of 100 dry battery cells tested to find the length of life produced the following result : $\bar{x} = 12\text{hrs}, \sigma = 3\text{hrs}$ Assuming the normal distribution what percentage of battery cell are expected to have life i) between 10 and 14 hrs, ii) less than 6 hrs (SNV : $z = 0$ to $z = 0.67, A = 0.2485, z = 0$ to $z = 2 A = 0.4772$) 4
2. A) Solve the following multiple choice questions.
 - i) In the least square method we use ----- to find the value of unknowns. 1
 - a) Regression equations b) Normal equations
 - c) General equations d) Auxiliary equations.
 - ii) The method of ----- is the most appropriate method to fit a unique curve from the given data. 1
 - a) Squares of square b) Sum of square
 - c) Least square d) None of these
- B) Solve **any two** questions of the following 4
 - i) fit a second degree parabola $y = a + bx^2$ for the following data

x	0	1	2	3
y	2	4	10	15

3) Solve any two questions of the following.

i) Find by Taylor's series method the value of y at $x = 0.1$ and $x = 0.2$ to five decimal places from $y' = x^2y - 1$, $y(0) = 1$ 4

ii) Using the Picard's method solve the following differential equations $\frac{dy}{dx} = 1 + xy$, $y = 1$, when $x = 0$ upto y^3 4

iii) Compute $y(0.2)$ given $\frac{dy}{dx} = x + y$, $y(0) = 1$ by using fourth order Range-Kutta method. 4

6. A) Solve the following multiple choice questions.

i) What is linear programming? 1

- a) Constrained optimization technique
- b) Technique for economic allocation of limited resources
- c) Mathematical technique
- d) All of the mentioned.

ii) The best use of linear programming technique is to find the optimal use of ----- 1

- a) Money
- b) Manpower
- c) Machine
- d) All the mentioned.

B) Solve any one questions of the following.

i) Two tailor A & B earn Rs. 150 and Rs. 200 per day respectively. A can stitch 6 shirt and 4 pants per day while B can stitch 10 shirt and 4 pants per day, form LPP minimize the labour cost to produce at least 60 shirts and 32 pants. 8

ii) Solve the following L. P. P graphically 8

$\max Z = 3x + 2y$
Subject to constraints,

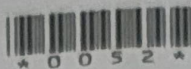
$$x + 2y \leq 10$$

$$3x + y \leq 15$$

$$x \geq 0, y \geq 0$$

B.Tech. First Year (Second Semester) Electrical Engineering / Mechanical Engineering
Engineering Mechanics
(EE24204 / ME24104)

Total Pages : 2



Time : One & Half Hour

Max. Marks : 30

Instructions to Candidates :

1. Draw neat and labeled diagram whenever necessary.
2. Due credit will be given to neatness and adequate dimension.
3. Assume suitable data whenever necessary.

1. A) Solve the following multiple-choice question/s 1
- i) Effect of Force on a body depends upon. 1
 - a) Magnitude
 - b) Direction
 - c) Position or line of action
 - d) All of the above
 - ii) A number of forces acting at a point will be in equilibrium if
 - a) Their total sum is zero
 - b) Two resolved parts in two directions at right angles are equal
 - c) Sum of resolved parts in any two perpendicular directions are both zero
 - d) All of them are inclined equally
- B) Solve **any two** questions of the following. 4
- i) Define equilibrium and state conditions of equilibrium. 4
 - ii) Determine the Resultant from the following system of forces
 - a) 20N inclined at 30° towards North-East.
 - b) 20N towards North.
 - c) 30N towards North-West.
 - d) 35N inclined at 40° towards south-west.
 - e) 50N inclined at 50° towards East-South.
 - iii) Determine reactions developed at supports in the beam as shown in Fig. 1 4

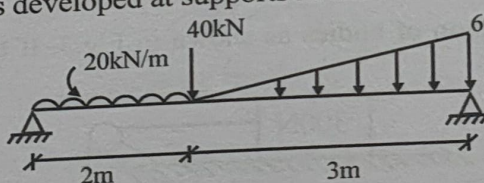


Fig. 1

2. A) Solve the following multiple-choice question/s 1
- i) ----- Trusses lie on a plane. 1
 - a) Planar
 - b) 2D
 - c) Linear
 - d) 3D
 - ii) ----- is known as the motion that would take place under the applied force if friction were absent. 1
 - a) Impending motion
 - b) Constrained motion
 - c) Unconstrained motion
 - d) Linear motion

P.T.O.

B) Solve **any two** questions of the following.

- i) State assumptions made in the analysis of truss. 4
- ii) Derive the relation between tension in tight side and slack side in belt friction. 4
- iii) For the block as shown in Fig. 2. Calculate the force P required to just move it up the plane take $\mu = 0.25$ 4

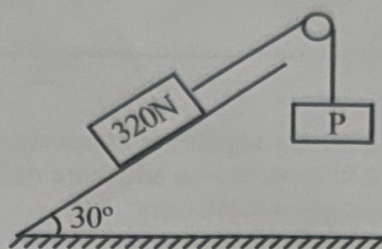


Fig. 2

3. A) Solve the following multiple-choice question/s.

- i) What is negative acceleration known as? 1
 - a) Deceleration
 - b) Acceleration
 - c) Escalation
 - d) Relaxation
- ii) D'Alembert's principle is used for which of the following? 1
 - a) Change static problem into a dynamic problem
 - b) Change dynamic problem to static problem
 - c) To calculate moment of inertia of rigid bodies
 - d) To calculate angular momentum of a system of masses

B) Solve **any two** questions of the following.

- i) An automobile starting from rest speed up to 40m/sec with constant acceleration of 4m/sec^2 , running at the speed for some time & finally comes to rest with retardation of 5m/sec^2 if total distance travelled is 1200m. Find the total time required. 4
- ii) A cricket ball through by the fielder from the height of 2m at an angle of 30° with horizontal at an initial velocity of 20m/sec. Its hits the wickets at the height of 0.5m from the ground. How far was the fielder from the wickets? 4
- iii) Determine Acceleration of bodies as shown in Fig.3. If the value of μ is 0.2. Find all contact faces. 4

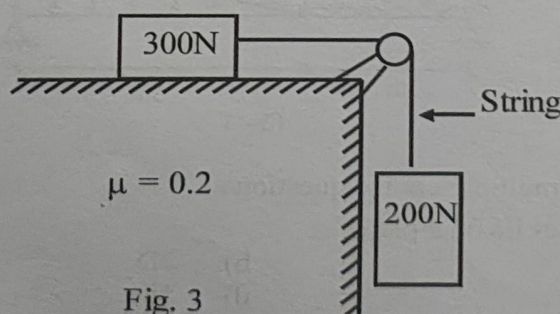


Fig. 3

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B.Tech. First Year (Second Semester) Civil Engineering
Basic of Electrical Engineering
(CE24204)

Total Pages : 2

Time : One & Half Hour



Max. Marks : 30

Instructions to Candidates :

1. All questions are compulsory.
2. Assume suitable data where ever necessary.
3. Draw neat and labeled diagram.

1. A) Solve the following multiple choice question/s.

- i) Which of the following equations correctly represents Kirchhoff's Voltage Law (KVL)? 1
 - a) The algebraic sum of all voltages in a closed loop is equal to zero
 - b) The algebraic sum of all currents entering a junction is zero
 - c) Voltage across each component in a parallel circuit is different
 - d) The total resistance in a series circuit is given by the sum of individual resistances
- ii) Which of the following statements about electrical resistance is true? 1
 - a) Resistance increases with an increase in temperature for all materials
 - b) Resistance is always directly proportional to conductivity
 - c) Resistance depends on the material, length, and cross-sectional area of a conductor
 - d) Resistance is inversely proportional to resistivity

B) Solve any two questions of the following.

- i) A toroidal coil has 500 turns and carries a current of 4A. If the mean radius of the toroid is 20 cm and the core has a relative permeability of 1000, find the magnetic field inside the toroid. (Use $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$) 4
- ii) Compare Magnetic circuit with Electric Circuit. 4
- iii) Discuss the principles of electromagnetic induction and differentiate between self-inductance and mutual inductance. 4

2. A) Solve the following multiple choice question/s.

- i) What is the relationship between the RMS value and the peak value of a sinusoidal waveform? 1
 - a) RMS value = Peak value
 - b) RMS value = Peak value / 2
 - c) RMS value = Peak value / $\sqrt{2}$
 - d) RMS value = Peak value $\times \sqrt{2}$

ii) In a star (Y) connection of a three-phase system, what is the relationship between phase voltage and line voltage?

- a) Phase voltage = Line voltage
- b) Phase voltage = Line voltage $\times \sqrt{3}$
- c) Phase voltage = Line voltage $/ \sqrt{3}$
- d) Phase voltage = Line voltage $/ 3$

B) Solve **any two** questions of the following.

- i) Explain the concept of root mean square (RMS) value in the context of alternating current (AC) circuits and discuss its significance in determining the effective value of AC voltage or current. 4
- ii) Describe the generation of three-phase AC supply, including the principles involved in producing three phase voltage and the significance of phase sequence in ensuring proper operation of three-phase systems. 4
- iii) Derive relationship between V_L and V_{ph} , I_L and I_{ph} for 3- phase Delta connected load. 4

3. A) Solve the following multiple choice question/s.

- i) Which type of wire is best suited for underground cabling to ensure durability and insulation? 1
 - a) PVC Insulated Copper Wire
 - b) Aluminum Conductor Cable
 - c) Armoured Cable
 - d) Twin Flexible Wire
- ii) What is the primary function of earthing in electrical installations? 1
 - a) To provide a path for the flow of excessive current during faults
 - b) To regulate the voltage levels in the electrical system
 - c) To increase the resistance of the electrical circuit
 - d) To decrease the power consumption of the electrical appliances

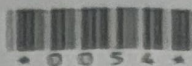
B) Solve **any two** questions of the following.

- i) A residential building has the following electrical loads: 5 LED bulbs of 15W each, 3 fans of 80W each and 2 air conditioners of 1.5 kW each. If all appliances run for 5 hours daily, calculate the total monthly energy consumption and the electricity bill at a tariff rate of ₹6 per unit. 4
- ii) Explain the working of a Switch Fuse Unit (SFU) and justify why it is used in industrial applications instead of a regular fuse. 4
- iii) Explain Pipe type of earthing. 4

B.Tech. First Year (Second Semester) Civil Engineering
Basics of Civil Engineering
(CE24205)

Total Pages : 2

Time : One & Half Hour



Max. Marks : 30

Instructions to Candidates :

1. Illustrate answer with suitable examples and diagram.
2. Marks for each question are indicated next to them.
3. Use only blue or black ink.

1. A) Solve the following multiple choice question/s

- | | |
|---|---|
| i) ----- is not the Branch of Civil engineering. | 1 |
| a) Environmental Engineering b) Geotechnical Engineering | |
| c) Social Engineering d) Structural Engineering | |
| ii) Highways are constructed for the reason -----. | 1 |
| a) To connect Cities & villages b) Highway is the quick option | |
| c) Highway reduces the travel time d) All the above | |

B) Solve any two questions of the following.

- | | |
|---|---|
| i) Write and explain different sources of water. | 4 |
| ii) What is the role of transportation engineering in national development. | 4 |
| iii) Give the classification of surveying. | 4 |

2. A) Solve the following multiple choice question/s.

- | | |
|--|---|
| i) Characteristics of Black cotton soil is to----- | 1 |
| a) Shrink b) Swell | |
| c) Expand d) All the above | |
| ii) In case of load bearing structure , mainly load to the foundation is transferred through---- | 1 |
| a) Columns b) Wall | |
| c) Parapet d) Non of above | |

B) Solve any two questions of the following.

- | | |
|---|---|
| i) What are the different types of loads acting on a building? Explain any one in detail. | 4 |
| ii) Explain: Pile foundation. | 4 |
| iii) Define and explain superstructure. | 4 |

3. A) Solve the following multiple choice question/s.

- i) One bag of Cement contains ----- Kg. of cement 1
a) 60 b) 50
c) 25 d) None of the above

- ii) Thickness of one brick thick wall is ----- 1
a) 430 mm b) 230 mm
c) 360 mm d) 100 mm

B) Solve **any two** questions of the following.

- i) Explain in detail: Bricks 4
ii) What is reinforced cement concrete? Also write uses of RCC. 4
iii) Explain: Eco friendly materials used in construction. 4

**B.Tech. First Year (Second Semester) Mechanical Engineering
Manufacturing Technology
(ME24205)**

Total Pages : 2



Time : One & Half Hour

Max. Marks : 30

Instructions to Candidates :

1. Due credit will be given to neatness and adequate dimensions.
2. Illustrate your answer necessary with help of neat sketches.
3. Use of pen Blue/Black ink/refill only for writing the Answer book.

1. A) Solve the following multiple choice questions.

- | | |
|--|--------------------------------|
| i) In resistance welding the electrode material is made of ----- | 1 |
| a) Carbon steel | b) Stainless steel |
| c) Copper | d) High speed steel |
| 1 | |
| ii) Seam, welding is ----- | |
| a) Arc welding | b) Multi spot welding |
| c) Continuous welding | d) Used for forming sound bars |

B) Solve **any two** questions of the following.

- | | |
|---|---|
| i) Describe the working principles of welding, soldering, and brazing with suitable applications. | 4 |
| ii) Explain the working principle of Submerged Arc Welding SAW and mention its advantages | 4 |
| iii) Differences between neutral, carburizing, and oxidizing flames in oxy-acetylene welding. | 4 |

2. A) Solve the following multiple choice questions.

- | | |
|---|------------------------|
| i) Which of the following is NOT a plastic manufacturing process? | 1 |
| a) Blow Molding | b) Compression Molding |
| c) Sand Casting | d) Rotational Molding |
| 1 | |
| ii) Thermosetting plastics have -----. | |
| a) 1-Degree bond | b) 2-Degree bond |
| c) 3-Degree bond | d) 0-Degree bond |

B) Solve **any two** questions of the following.

- | | |
|---|---|
| i) Explain the different types of plastic materials used in manufacturing and their properties. | 4 |
|---|---|

ii) Describe the rotational molding process and its significance in producing hollow plastic components. 4

iii) Explain the difference between compression molding and thermoforming. 4

3. A) Solve the following multiple choice questions.

- i) In hot working, the metal is deformed at a temperature. 1
- Below its melting point but above recrystallization temperature.
 - Above its melting point
 - At room temperature
 - Below recrystallization temperature

- ii) Which of the following is a hot extrusion process? 1
- Direct extrusion
 - Stamping
 - Coining
 - Blanking

B) Solve **any two** questions of the following.

i) Explain backward extrusion with a sketch. 4

ii) Explain wire drawing with a sketch. 4

iii) Describe the defects in rolled parts. 4

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B.Tech. First Year (First Semester) Electrical Engineering / Civil Engineering /
Mechanical Engineering / Electronics & Telecommunication Engineering
Matrices and Differential Equations
(EE24101 / CE24101 / ME24101 / ET24101)

Total Pages : 4

Time : Three Hours



Max. Marks : 60

Instructions to Candidates :

1. Assume suitable data wherever necessary.
2. Use of pen Blue/Black ink/refill only for writing the Answer book.
3. Use of calculator is allowed.

1. A) Solve the following multiple-choice question/s.

i) Which of the following is the characteristics equation of matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} ?$$

a) $\lambda^2 + 2\lambda + 1 = 0$

b) $\lambda^2 + \lambda = 0$

c) $\lambda^2 - 5 = 0$

d) $\lambda^2 - 5\lambda + 3 = 0$

ii) What are the Eigen values for matrix

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} ?$$

a) 2, 1, 1

b) -2, 3, 6

c) -1, 3, 2

d) 5, -3, -2

B) Solve **any two** questions of the following.

i) Determine rank of the matrix

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

ii) Apply matrix method to check consistency & hence find solution of system:

$$x + 2y + z = 3$$

$$2x + 3y + 2z = 5$$

$$3x - 5y + 5z = 2$$

$$3x + 9y - z = 4$$

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iii) Use Cayley-Hamilton theorem to find Λ^{-1} , if

$$\Lambda = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

2. A) Solve the following multiple-choice question/s.

- i) Function $f(t)$ can be represented by a Fourier series if it is
- a) Periodic
 - b) Continuous
 - c) Discontinuous
 - d) Differentiable

- ii) Find b_n of the function $f(x) = x^2 - 2$ in interval $(-2, 2)$
- a) 0
 - b) $-2/3$
 - c) $-1/3$
 - d) $-4/3$

B) Solve **any one** question of the following.

- i) Obtain the Fourier series for $f(x) = x^2$ in $(0, 2\pi)$.

- ii) Find the constant term and coefficient of first Sine and Cosine terms in Fourier Series of y , where y is given in following table:

x	0	1	2	3	4	5
y	9	18	24	28	26	20

3. A) Solve the following multiple-choice question/s.

- i) Differential equation of the form $Mdx + Ndy = 0$ is said to be Exact, if

- a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- b) $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$
- c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
- d) $\frac{\partial M}{\partial x} \neq \frac{\partial N}{\partial y}$

- ii) Solution of $\frac{dy}{dx} = \frac{y}{x}$ is -----

- a) $\frac{1}{y} = x$
- b) $\frac{x^2}{y} = c$
- c) $\frac{c}{y} = x$
- d) $\frac{y}{x} = c$

B) Solve **any two** questions of the following.

- i) Solve $\frac{dy}{dx} = (4x + y + 1)^2$

ii) Solve $\frac{dy}{dx} + \left(\frac{4x}{1+x^2}\right)y = \frac{1}{1+x^2}$ 4

iii) Solve $(y \cos x + \sin y + y) dx + (\sin x + x \cos y + x) dy = 0$. 4

4. A) Solve the following multiple-choice question/s. 1

i) Solution of $(p+q)(z-xp-yq)=1$ is

a) $z = xa - yb - \frac{1}{a+b}$

b) $z = xa + yb + (a+b)$

c) $z = xa + yb + \frac{1}{a+b}$

d) $z = xa - yb - (a+b)$

ii) Lagrange's partial differential equation is of the form 1

a) $Pp + Qq = R$

b) $f(p, q) = 0$

c) $f(p, q, z) = 0$

d) $f(p, q, x, y) = 0$

B) Solve any two questions of the following.

i) Solve $pq = p+q$. 4

ii) Solve $p^2 + q^2 = z$. 4

iii) Solve $(y-z)p + (z-x)q = x-y$. 4

5. A) Solve the following multiple-choice question/s.

i) Identify value of $\cos 0 + i \sin 0$. 1

a) 1

b) -1

c) i

d) -i

ii) $(\cos \pi + i \sin \pi)$ can also be written as 1

a) $\cos(2k+1)\pi + i \sin(2k+1)\pi$

b) $\cos 2k\pi + i \sin 2k\pi$

c) $\cos 0 + i \sin 0$

d) None of these

B) Solve any two questions of the following.

i) Show that 4

$$(a+ib)^{m/n} + (a-ib)^{m/n}$$

$$= 2(a^2 + b^2)^{m/2n} \cos\left(\frac{m}{n} \tan^{-1} \frac{b}{a}\right)$$

ii) Show that $\sinh^{-1} x = \log\left(x + \sqrt{x^2 + 1}\right)$ 4

- iii) Find the roots of the equation $x^9 - x^5 + x^4 - 1 = 0$.
6. A) Solve the following multiple-choice question/s.

- i) To find a tabulated value near the ending of the table, use
- Newton's forward formula
 - Newton's backward formula
 - Newton's divided difference formula
 - Extrapolation
- ii) Interpolation is a method of
- Interrelating
 - Integrating
 - Estimating
 - Combining

- B) Solve **any one** question of the following.

- i) Apply Lagrange's interpolation formula to find the value of y , when $x = 2$.
- | | | | | |
|---|-----|---|---|----|
| x | 0 | 1 | 3 | 4 |
| y | -12 | 0 | 6 | 12 |

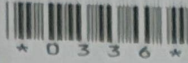
- ii) Estimate the value of $f(3)$ using Newton's forward interpolation formula for the data:
- | | | | | |
|------|------|------|-------|-------|
| x | 1 | 6 | 11 | 16 |
| f(x) | 3.69 | 8.69 | 15.69 | 26.69 |

B.Tech. First Year (First Semester) Electrical Engineering / Mechanical Engineering
Basic Electrical Engineering

(EE24103 / ME24103) ✓

Total Pages : 4

Time : Three Hours



Max. Marks : 60

Instructions to Candidates :

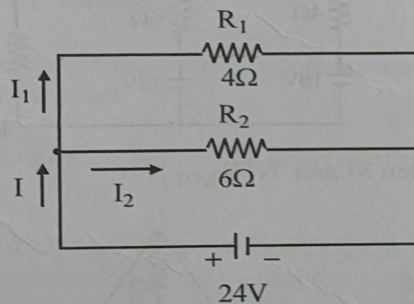
1. All question carry equal marks.
2. All questions are compulsory.
3. Assume suitable data wherever necessary.
4. Illustrate your answer with the help of neat sketches.
5. Draw neat and labeled diagram.

1. A) Solve the following multiple choice question/s.

- i) Kirchhoff's Current Law is based on 1
 - a) The charge can be accumulated at the node
 - b) Charge cannot be accumulated at the node
 - c) Energy is stored at the node
 - d) Depending on the circuit charge can be accumulated at the circuit
- ii) A closed path made by several branches of the network is known as 1
 - a) Branch
 - b) Loop
 - c) Circuit
 - d) Junction

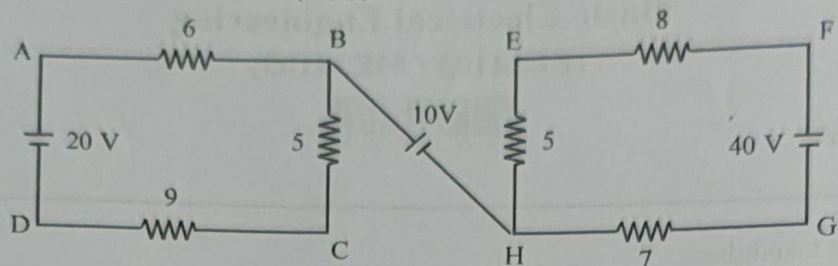
B) Solve any two questions of the following.

- i) Calculate the equivalent resistance in the following circuit and also find the current I , I_1 and I_2 in the given circuit. 4



- ii) Discuss the following terms: 4
 - a) Ohms law
 - b) Kirchhoff's Laws
 - c) Electrical Power
 - d) Electrical potential

iii) Find voltage between C and E (V_{CE}) and voltage between A and G (V_{AG}).



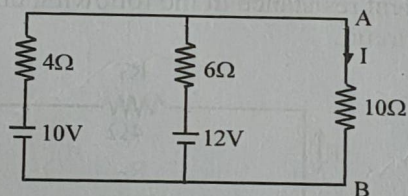
2. A) Solve the following multiple choice question/s.

- i) To apply the superposition theorem required circuit must be -----
 - a) Linear circuit
 - b) Bilateral Circuit
 - c) Having more than one source
 - d) All of the above

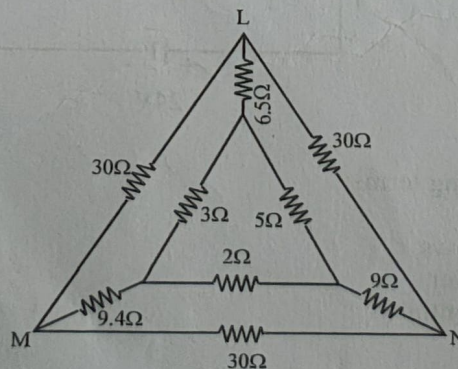
- ii) Thevenin resistance is found by -----
 - a) Shorting all voltage sources
 - b) Opening all current sources
 - c) Shorting all voltage sources and opening all current sources
 - d) Opening all voltage sources and shorting all current sources

B) Solve **any two** questions of the following.

- i) State and Explain Superposition theorem with example.
- ii) Obtain Thevenin's equivalent circuit with respect to the terminals of AB of the network shown in the circuit and find I.



iii) Find resistance between M and N (R_{MN}).



3. A) Solve the following multiple choice question/s.

- | | | |
|--|-------------------|---|
| i) Reciprocal of reluctance is? | | |
| a) Reluctivity | b) Conductivity | 1 |
| c) Permeance | d) Susceptibility | |
| | | |
| ii) The mutual inductance (M) between the two coils is 3H. The self inductance of the two coils are 4H and 9H respectively. The coefficient of coupling is | | 1 |
| a) 0.3 | b) 0.4 | |
| c) 0.5 | d) 0.6 | |

B) Solve **any two** questions of the following.

- | | |
|---|---|
| i) Compare the Electric circuit with magnetic circuit. | 4 |
| ii) A flux density of 1.2 wb/m^2 is required in the 2mm air gap of electromagnet having iron path 1m long. Calculate the mmf required. Assuming the relative permeability of iron as 1500. Neglect leakage. | 4 |
| iii) Discuss the concept of mutual induction and mutual induced emf. Also derive the expression for the mutual inductance. | 4 |

4. A) Solve the following multiple choice question/s.

- | | |
|--|---------------------------------|
| i) How many cycles will an AC signal make in 2 seconds if its frequency is 100 Hz? | 1 |
| a) 50 | b) 100 |
| c) 150 | d) 200 |
| | |
| ii) The j operator is symbol used for an operation to indicate ----- | 1 |
| a) Clock wise direction | b) Counter clock wise direction |
| c) 180° Clock wise direction | d) 270° Clock wise direction |

B) Solve **any two** question of the following.

- | | |
|---|---|
| i) Derive the expression of current, circuit impedance and power for pure Capacitor with waveform and Phasor diagram. | 4 |
| ii) Derive the relation between average value of an alternating current with maximum value of alternating current. | 4 |
| iii) The pure inductance of 318 mH is connected in series with a pure resistance of 75Ω . The circuit is supplied from 50Hz source and the voltage across 75Ω resistor is found to be 150V. Calculate the supply voltage and phase angle. | 4 |

5. A) Solve the following multiple choice question/s.

- | | |
|---|---|
| i) In balance three phase load ----- | 1 |
| a) Magnitude of Voltage is same | |
| b) Magnitude of current is same | |
| c) Magnitude of Voltage is same with 120° phase difference | |
| d) Magnitude of current is same with 120° phase difference | |

- ii) The equation of three phase power is given by
- | | |
|---------------------------------|---------------------------------|
| a) $\sqrt{3} V_p I_p \cos \Phi$ | b) $\sqrt{3} V_L I_L \cos \Phi$ |
| c) $\sqrt{3} V_L I_L \cos \Phi$ | d) $V I \cos \Phi$ |

B) Solve **any two** questions of the following.

- i) What are the advantages of a three phase supply system over a single phase supply system? 4
- ii) Derive and explain relationship between line and phase quantities for 3-phase star connected load with circuit diagram and phasor diagram. 4
- iii) Three identical coils, each of resistance 10ohm and inductance 42mH are connected (a) in star and (b) in delta to a 415V, 50 Hz, 3-phase supply. Determine the total power dissipated in each case. 4

6. A) Solve the following multiple choice question/s.

- i) The operating time of the fuse is ----- 1
- | | |
|------------|--------------|
| a) 0.2 Sec | b) 0.002 Sec |
| c) 0.1 Sec | d) 0.001 Sec |
- ii) The maximum short circuit rating of MCB is ----- 1
- | | |
|-------------------|----------------------|
| a) 100KA at 240 V | b) 200KA, 240 V |
| c) 16KA, 240 V | d) None of the above |

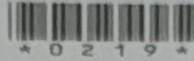
B) Solve **any two** questions of the following.

- i) Discuss the advantages and disadvantages of fuse. 4
- ii) Explain construction, working and operation of MCB. 4
- iii) Explain Pipe type Earthing. 4

B.Tech. First Year (First Semester) Civil Engineering
Engineering Mechanics
(CE103)

Total Pages : 5

Time : Three Hours



Max. Marks : 60

Instructions to Candidates:

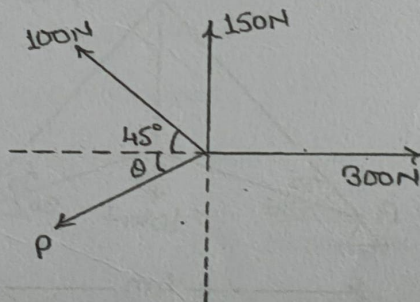
1. All question carry equal marks.
2. Due credit will be given to neatness and adequate Dimensions.
3. Assume suitable data wherever necessary.
4. Use of pen Blue/Black ink/refill only for writing in Answer book.

1. a) Solve the following multiple-choice questions.

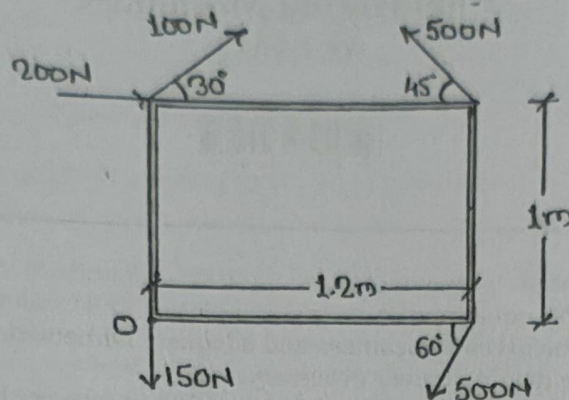
- | | | |
|-----|--|---|
| i) | A system of three forces acts on a body & keeps it in equilibrium. The forces need to be ----- | 1 |
| a) | Coplanar only | |
| b) | Concurrent only | |
| c) | Coplanar as well as concurrent | |
| d) | Coplanar but may or may not be concurrent | |
| ii) | The process of splitting a single force into its components is known as ----- | 1 |
| a) | Resultant | |
| b) | Resolution of force | |
| c) | Composition of force | |
| d) | Moment of force | |

b) Solve **any two** questions of the following.

- | | | |
|-----|---|---|
| i) | State & explain Law of parallelogram of force. | 4 |
| ii) | For the system of Coplanar concurrent forces system. If the resultant is zero, find the magnitude of force 'P'. | 4 |



iii) Determine the magnitude, direction & position of resultant force as shown in fig.



2. a) Solve the following multiple choice questions.

i) A framed structure is said to be perfect if the following correlation exists between the number of joints 'j' & the number of members 'm'.

a) $m = 2j - 3$

b) $m = 3j - 3$

c) $m = 2j - 1$

d) $m = j - 2$

ii) If the number of members are less than that of required for the perfect truss then it is thrown as -----

a) Stable truss

b) Unstable truss

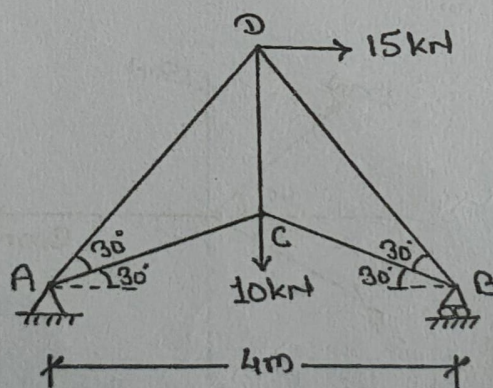
c) Perfect truss

d) Over perfect truss

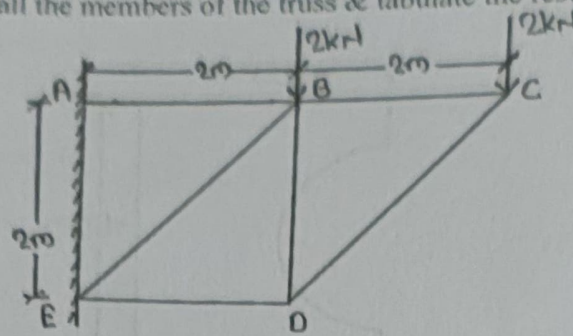
b) Solve **any two** questions from the following.

i) Describe the relationship between tension on tight side & slack side.

ii) The truss has been loaded & supported as shown in fig. find the forces in all the members of the truss.



iii) Find the forces in all the members of the truss & tabulate the results.



3. a) Solve the following multiple choice questions.

i) For the rectangular element with side b & h , the moment of inertia about its centroidal axis is -----

- a) $bh^3/3$
- b) $bh^3/12$
- c) $bh^2/6$
- d) $bh^3/36$

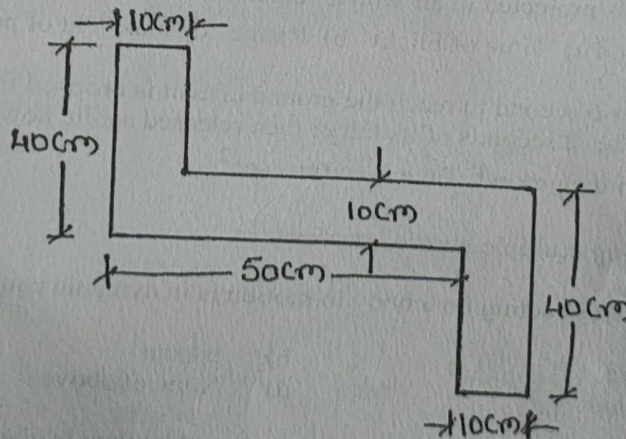
ii) The MI of a body does not depend upon -----

- a) Shape of the body
- b) Mass of body & its distribution within the body
- c) Axis of rotation of body
- d) Angular velocity of body

b) Solve **any two** questions from the following.

i) State & explain parallel axis theorem.

ii) Determine the MI about centroidal axis as shown in fig.



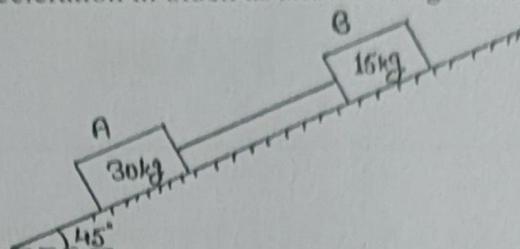
b) Solve **any two** questions of the following.

i) State & explain D'Alembert's principle.

4

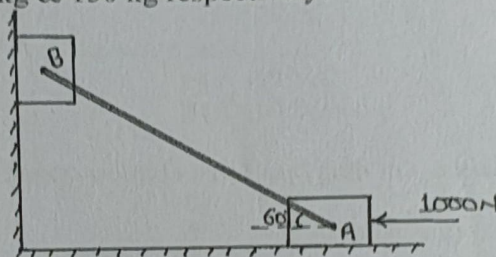
ii) Find the tension & acceleration in block as shown in fig. Take $\mu_A = 0.2$ & $\mu_B = 0.3$.

4



iii) Two rigid blocks A & B connected by massless rigid rod 2m long as shown in fig. If the system starts from rest at the given position by applying a force of 1000 N as shown. Calculate acceleration of two blocks. Assume surface to be frictionless. Masses of blocks A & B are 200 kg & 150 kg respectively.

4



6. a) Solve the following multiple choice questions.

i) When the bullet is fired from a gun, it recoils in the backward direction. It is due to -----

1

- | | |
|-----------------------------|---------------------------|
| a) Impulse | b) Inertia |
| c) Conservation of momentum | d) Conservation of energy |

ii) The work done on a body equals the change in ----- of the body.

1

- | | |
|---------------------|-------------------|
| a) Energy | b) Kinetic energy |
| c) Potential energy | d) Distance |

b) Solve **any two** from the following.

i) Explain Impulse-Momentum equation.

4

ii) Explain the terms:

4

- | | |
|------------------------|-------------------|
| a) Line of impact | b) Central impact |
| c) Coe. Of restitution | d) Direct impact |

iii) A ball of mass 2kg moving with a velocity of 3m/sec impinges directly on a ball of mass 4kg at rest. After impact, the 2kg mass ball comes to rest. Determine the velocity of 4kg ball after striking & the coefficient of restitution between the two balls.

4

**B.Tech. First Year (First Semester) Electronics Engineering / Civil Engineering /
Mechanical Engineering
Engineering Graphics
(EE24104 / CE24104 / ME24104)**

Total Pages : 4



Time : Three Hours

Max. Marks : 60

Instructions to Candidates :

1. All question carry equal marks.
2. Due credit will be given to neatness and adequate Dimensions.
3. Assume suitable data wherever necessary.
4. Pertain the construction lines.
5. Illustrate your answer with the help of neat sketches.
6. Use of scientific calculator is permitted.
7. Use of pen Blue/Black ink/refill only for writing in Answer book.
8. Use of drawing instruments is permitted.

1. A) Solve the following multiple-choice questions.

- i) When a bullet is shot in air the path traversed by the bullet is called 1
- | | |
|---------------|--------------|
| a) Ellipse | b) Parabola |
| c) Semicircle | d) Hyperbola |

- ii) Name the curve which has zero eccentricity. 1

- | | |
|--------------|-------------|
| a) Ellipse | b) Parabola |
| c) Hyperbola | d) Circle |

B) Solve any one questions of the following.

- i) Construct rectangular hyperbola, given the position of point on it, 40 mm and 25 mm from the asymptotes OX and OY respectively. 8

- ii) In a simple slider crank mechanism, the connecting rod is 120 mm and crank is 30 mm in length. The other end point of connecting rod on the slider moves a straight line passing through centre of crank rotation. Construct the locus of mid-point of connecting rod. 8

2. A) Solve the following multiple-choice question/s.

- i) If the front view of a line is parallel to the xy its true length is shown in 1
- | | |
|---------------|------------------------------|
| a) Front view | b) Top view |
| c) Side view | d) Both front and side views |

- ii) If the top view of a plane is a rhombus the object may be 1
- | | |
|------------------|-----------------|
| a) Parallelogram | b) Square |
| c) Octagon | d) Any of these |

B) Solve **any one** questions of the following.

- i) a) The top view of a line measures 60 mm. The line is parallel to the V.P. and inclined at 45° to the H.P. One end of the line is 30 mm in front of the V.P. and lies on the H.P. Construct its projections and determine its true length. 1
- b) A 100 mm long line is parallel to and 40 mm below the H.P. its two ends are 25 mm and 50 mm behind the V.P. respectively. Construct its projections and find its inclination with the V.P. 4
- ii) A thin pentagonal plate of side 30 mm is placed on a side in the H.P. inclined at 30° to the V.P. The corner opposite to that side is 35 mm above the H.P. Construct its projections. 8

3. A) Solve the following multiple-choice question/s.

- i) In a pyramid triangular faces meeting at a point is called as 1
 - a) Center point
 - b) Midpoint
 - c) Edge point
 - d) Vertex
- ii) If a solid is cut by a section plane and it is projected to a plane parallel to it, the shape of the section obtained is 1
 - a) Sectional view
 - b) Apparent shape
 - c) True shape
 - d) None of the above

B) Solve **any one** questions of the following.

- i) A right circular cone of base circle diameter 50 mm and axis 70 mm long is resting on its generator on HP, such that the top view of the axis is inclined at an angle of 30° to the VP. Construct its projections. 8
- ii) A square pyramid, base 40 mm and axis 65 mm long, has its base on the HP and all the edges of the base equally inclined to the VP. It is cut by a section plane, perpendicular to the VP, inclined at 45° to the HP and bisecting the axis. Construct its sectional top view, FV and true shape of the section. 8

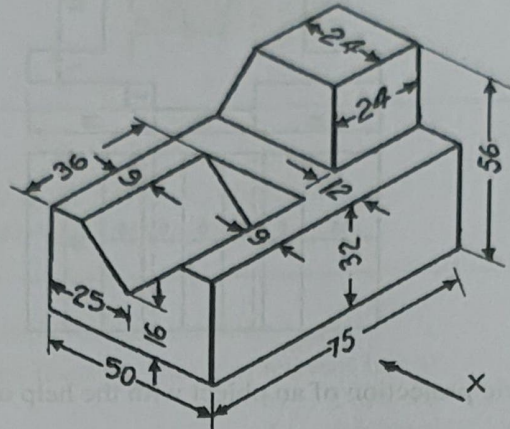
4. A) Solve the following multiple-choice questions.

- i) In orthographic projection, visual rays or lines of sight for given view are ----- to each others. 1
 - a) perpendicular
 - b) normal
 - c) oblique
 - d) parallel
- ii) For the third angle projection method, which of the following is correct? 1
 - a) Observer – Object – Plane
 - b) Observer – Plane – object
 - c) (a) and (b) both
 - d) None of above

B) Solve any one questions of the following.

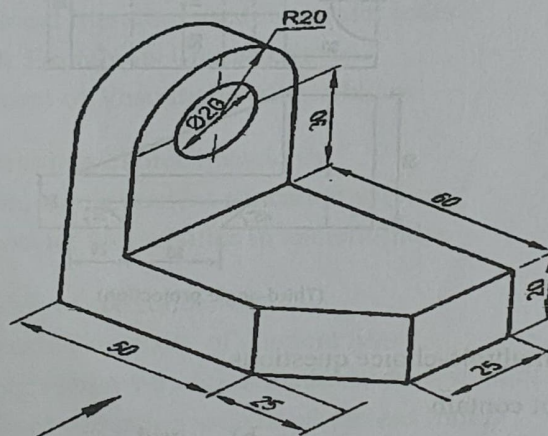
- i) Fig. No. 1 shows a pictorial view of an object, Construct (a) Front View (b) Top View and (c) Left Hand Side View by First Angle Projection Method.

8



- ii) Fig. No. 2 shows a pictorial view of an object, Construct (a) Front View (b) Top View and (c) Right Hand Side View by Third Angle Projection Method.

8



5. A) Solve the following multiple-choice question/s.

- i) The isometric axes are inclined at ----- to each other.

1

- a) 30° b) 60°
c) 90° d) 120°

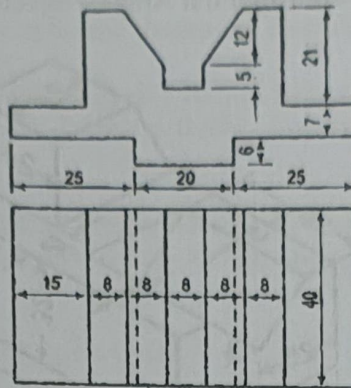
- ii) True (natural) scale makes ----- angle with base line and isometric scale makes ----- angle with base line.

1

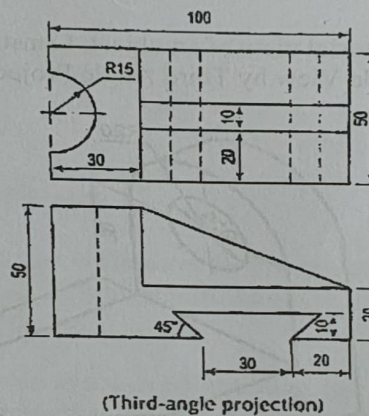
- a) $30^\circ, 45^\circ$ b) $45^\circ, 30^\circ$
c) $30^\circ, 60^\circ$ d) $60^\circ, 30^\circ$

B) Solve **any one** questions of the following.

- i) Construct an isometric view of an object with the help of given views as shown in Fig. No. 3.



- ii) Construct isometric projection of an object with the help of given views as shown in Fig. No. 4.



6. A) Solve the following multiple-choice questions.

- i) Status bar do not contain
a) snap
b) grid
c) erase
d) polar
- ii) When using the TRIM commands, which of the following do you select first?
a) Cutting edges
b) The object to be trimmed
c) Everything
d) Nothing

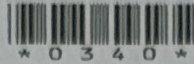
B) Solve **any two** questions of the following.

- i) Explain the use of chamfer and fillet.
- ii) Explain the different coordinate systems used in CAD software.
- iii) Describe some common editing commands in CAD software.

**B.Tech. First Year (First Semester) Electrical Engineering / Civil Engineering /
Mechanical Engineering / Electronics & Telecommunication Engineering
Indian Knowledge System - EE24105 / CE24105 / ME24105 / ET24105**

Total Pages : 1

Time : One & Half Hour



Max. Marks : 30

Instructions to Candidates :

1. Question No. 1 A, 2 A, & 3A are Compulsory.
2. Due marks will be given to neatness.

1. A) Solve the following multiple-choice question/s.
 - i) Which is the earliest literacy source of ancient India? 1
 - a) Vedas b) Puranas
 - c) Smritits d) Brahmana
 - ii) Enlist any two sanskaras names as per ancient India. 1
- B) Solve **any two** questions of the following.
 - i) Explain foundational literature used in ancient India. 4
 - ii) Define Culture & Explain its Characteristics. 4
 - iii) Explain the Concept of Vasudhaive Kutumbkam. 4
2. A) Solve the following multiple-choice question/s.
 - i) Enlist various subjects in ancient education system. 1
 - ii) Enlist any two ancient Universities in ancient India. 1
- B) Solve **any two** questions of the following.
 - i) Explain the Nalanda University of Ancient time. 4
 - ii) Explain the salient features traditional education system of ancient India. 4
 - iii) Explain the mode of education system of ancient India. 4
3. A) Solve the following multiple-choice question/s.
 - i) Who is the father of Surgery? 1
 - ii) Who is the Inventor of Zero? 1
- B) Solve **any two** questions of the following.
 - i) Explain the India's contribution in Water management system. 4
 - ii) Explain the contribution of ancient India in the field of Ship Building. 4
 - iii) Explain the various mathematician's contribution of ancient India in the field of Mathematics. 4

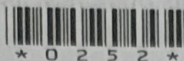
B.Tech. First Year (Second Semester) Civil Engineering / Electrical Engineering / Electronics & Telecommunication Engineering / Mechanical Engineering / Artificial Intelligence & Data Science / Computer Science & Engineering / Machine Learning

Professional Communication Skills

(CE24206 / EE24206 / ET24206 / ME24206 / AI24106 / CS24106 / ML24106)

Total Pages : 3

Time : One & Half Hour



Max. Marks : 30

Instructions to Candidates :

1. Write neat and clean using blue or black pen
2. All questions carry marks as indicated.
3. All Questions are Compulsory (Q1A, Q2A, Q3A, Q1Bi)
4. Questions have options (Q1B ii OR Q1 Biii) and (Q2B and Q3B have options)

1. A) Solve the following multiple choice question/s.

i) Synonyms

1. fostering

- a) Safeguarding
~~b) Ignoring~~

- ~~b) Neglecting~~
 d) Nurturing

2. Cease

- ~~a) Begin~~
 c) Create

- b) Stop
~~d) Dull~~

ii) Antonyms

1. foremost

- a) Hindmost
~~c) Disposed~~

- b) Unimportant
 d) Mature

2. mighty

- a) Forcible
 c) Weak

- ~~b) Forceful~~
 d) Tough

B) Solve any two questions of the following.

i) Solve the questions on Comprehension.

Mike and Morris lived in the same village. While Morris owned the largest jewellery shop in the village, Mike was a poor farmer. Both had large families with many sons, daughters-in-law and grandchildren. One fine day, Mike, tired of not being able to feed his family, decided to leave the village and move to the city where he was certain to earn enough to feed everyone. Along with his family, he left the village for the city. At night, they stopped under a large tree. There was a stream running nearby where they could freshen up themselves. He told his sons to clear the area below the tree, he told his wife to fetch water and he instructed his daughters in-law to make up the fire and started cutting wood from

the tree himself. They didn't know that in the branches of the tree, there was a thief hiding. He watched as Mike's family worked together and also noticed that they had nothing to cook. Mike's wife also thought the same and asked her husband, "Everything is ready but what shall we eat?" Mike raised his hands to heaven and said, "Don't worry. He is watching all of this from above. He will help us." The thief got worried as he had seen that the family was large and worked well together. Taking advantage of the fact that they did not know he was hiding in the branches, he decided to make a quick escape. He climbed down safely when they were not looking and ran for his life. But, he left behind the bundle of stolen jewels and money which dropped into Mike's lap. Mike opened it and jumped with joy when he saw the contents. The family gathered all their belongings and returned to the village. There was great excitement when they told everyone how they got rich. Morris thought that the tree was miraculous and this was a nice and quick way to earn some money. He ordered his family to pack some clothes and they set off as if on a journey. They also stopped under the same tree and Morris started commanding everyone as Mike had done. But no one in his family was willing to obey his orders. Being a rich family, they were used to having servants all around. So, the one who went to the river to fetch water enjoyed a nice bath. The one who went to get wood for fire went off to sleep. Morris's wife said, "Everything is ready but what shall we eat?" Morris raised his hands and said, "Don't worry. He is watching all of this from above. He will help us." As soon as he finished saying, the thief jumped down from the tree with a knife in hand. Seeing him, everyone started running around to save their lives. The thief stole everything they had and Morris and his family had to return to the village empty handed, having lost all their valuables that they had taken with them.

Question 1: Why did Mike and his family decide to rest under the thief's tree?

- ☒ A) Being a large family, they knew that they could easily defeat the thief
- ☐ B) It was a convenient spot for taking a halt at night
- ☐ C) There was a stream nearby and wood enough to build a house
- ☐ D) That was the only large tree that could shelter their large family

Question 2: Which of the following best describes Morris?

- ☒ A) He was a rich businessman
- ☐ B) He bullied his wife
- ☐ C) He paid his servants well
- ☐ D) He was greedy and imitated Mike

Question 3: What did Mike mean when he said "He is watching all this from above"?

- ☐ A) He had spotted the thief and wanted to scare him
- ☐ B) He was telling his wife to have faith in god
- ☒ C) It was just a warning for his family members to stick together
- ☐ D) He was begging the thief to help his family

Question 4: How did the fellow villagers react to Mike getting rich overnight?

- ☐ A) They were jealous of him
- ☒ B) They were very excited
- ☐ C) They followed his example
- ☐ D) They envied him

ii) Solve **any one** from below do as directed (Q1 B ii or Q1 B iii)

Do as directed

1. Everyone loves Zara..(Change the voice)
2. He will return back next week.)(Error identifications and rewrite correct)
3. Joya's father is ----- principal of our college.(fill articles)
4. ----- you open the gate please? (could, should)

4

iii) Do as directed

1. Sam had taken the medicines.(Change the voice)
2. The teacher of English(A) / gave him a home work (B) / to complete that (C) / by night positively(D) / No error (E).(Error identifications)
3. ----- Market will remain closed for two days.(fill articles)
4. She ----- surely help you. (will, must)

4

2. A) Solve the following multiple choice question/s.

- i) Which of these is the most important tool of communication?
- a) Posture
 - ☒ b) Language
 - c) Body language
 - d) Gestures

1

- ii) Which of these must be avoided for effective communication?
- a) Sharing of activity
 - b) Politeness
 - c) Ambiguity
 - d) Listening

1

B) Solve **any two** questions of the following.

- i) What are psychological barrier to communication explain with suitable example.
- ii) Explain dos and don't s of group discussions.
- iii) Draw and explain communication cycle.

4

4

4

3. A) Solve the following multiple choice question/s.

- i) What is the most effective way of closing an email?
- ☒ a) With a social message
 - b) With a clear and actionable message
 - c) Repeating the subject line
 - d) With a proverb

1

ii) What does BCC stand for?

- a) Big Cobra Coming
- ☒ b) Blind carbon copy
- c) Blind copy copied

1

B) Solve **any two** questions of the following.

- i) Write a paragraph for E vehicles..
- ii) Write an email to HR Linda.WilliamsHR@email.com of a company for internship availability.
- iii) Draft a complaint letter to swayam electronics for receiving defective washing machines for SK hotel.

4

4

4

S25/1006

B.Tech. First Year (First Semester) Civil Engineering / Mechanical Engineering
General Physics and Acoustics
(CE24102 / ME24102)

Pages : 3
 : Three Hours



Max. Marks : 60

2018

Instructions to Candidates :

1. Draw well labeled Diagram wherever necessary.
2. Mass of Electron $= 9.1 \times 10^{-31} \text{ kg}$;
3. Charge on Electron $= 1.6 \times 10^{-19} \text{ J}$;
4. Plank's Constant $h = 6.63 \times 10^{-34} \text{ J sec}$
5. Velocity of light $= 3 \times 10^8 \text{ m/s}$

A) Solve the following question/s.

- i) In an Intrinsic semiconductor, position of Fermi energy Level is ----- 1
 - a) In between C.B. and Donor level
 - b) In between V.B. and Acceptor level
 - c) Exactly at the center of C.B. and V.B.
 - d) Below V.B.
- ii) Which of the following is the majority charge carrier in P-type of the Semiconductor? 1
 - a) Electrons
 - b) Holes
 - c) Positive donor ions
 - d) Negative acceptor ions

B) Solve any two questions of the following.

- i) Define mobility and State the conductivity equations for Intrinsic and Extrinsic semiconductors. 4
- ii) With necessary circuit diagram Explain the reverse bias mechanism in PN-junction diode. 4
- iii) Differentiate between Zener Breakdown and Avalanche Breakdown. 4

2. A) Solve the following multiple choice questions.

- i) The De-Broglie's matter wave is generated by 1
 - a) Only charged particle
 - b) Any particle at rest
 - c) Any particle which is in motion
 - d) Only micro particles
- ii) Compton Effect is observed in ----- Element 1
 - a) Lighter
 - b) Heavier
 - c) Radioactive
 - d) Semiconductor

S25/1006

1

P.T.O.

B) Solve any two questions of the following.

- i) State any four properties and four applications of LASER.
- ii) Discuss the experimental arrangement of Compton effect experiment with the help of necessary diagram.
- iii) Calculate the De-Broglie's wavelength associated with proton moving with energy 1 Mev. (given: mass of Proton = 1.6×10^{-27} kg)

3. A) Solve the following questions.

- i) In CRO input signal is applied to -----
 - a) Time base Generator
 - b) Horizontal amplifier
 - c) Vertical Amplifier
 - d) Trigger circuit
- ii) In perpendicular magnetic field the electron traces ----- path.
 - a) Parabolic
 - b) Straight
 - c) Circular
 - d) Zig zag

B) Solve any two questions of the following.

- i) State and explain any four properties of positive rays.
- ii) Draw Block diagram of CRO and specify the working of Horizontal Amplifier.
- iii) An electron of velocity 2×10^7 m/s is circulating in a plane at right angles to uniform field of flux density 1.5 Wb/m^2 . Calculate the radius of electron orbit and period of revolution.

4. A) Solve the following question/s.

- i) The plane transmission grating is used to study the ----- phenomenon of light.
 - a) Diffraction
 - b) Interference
 - c) Polarisation
 - d) Scattering
- ii) In Newton's Ring experiments, the diameter of Bright rings is proportional to -----
 - a) Square root of Odd Natural numbers.
 - b) Natural Number
 - c) Even Natural Number
 - d) Square root of natural number

B) Solve any two questions of the following.

- i) Define Diffraction. Discuss the Fresnel class of diffraction of light with diagram.
- ii) Derive the equation for radius dark ring in case of Newton's ring experiment.

- iii) In Newton's ring are observed in reflection of light of W.L. $6000\text{\AA}$. The diameter of 9th dark ring is found to be 0.5 cm. Find the radius of curvature of lens and thickness of the film. 4

A) Solve the following multiple choice question.

- i) Which of the following statement is wrong regarding crystalline solids: 1
- Crystalline solids possess long range order
 - They have sharp melting point
 - The atoms are arranged in regular manner
 - The atoms are not arranged in orderly manner
- ii) The crystal structure is the result of two quantities; namely ----- 1
- Lattice + Basis = Crystal Structure
 - Lattice + Lattice points = Crystal Structure
 - Both a and b are correct
 - Both a and b are incorrect

B) Solve **any two** questions of the following.

- i) Explain in brief the monoclinic and Triclinic crystal system. 4
- ii) Define density of packing and determine the packing fraction of Simple cubic structure. 4
- iii) Explain in brief the terms Axes of Symmetry and planes of symmetry of a cubic crystal. 4

6. A) Solve the following multiple choice question.

- i) The coefficient of viscosity of Fluid has unit ----- 1
- N sec/m^2
 - N/sec.m^2
 - Nsec.m^2
 - N/m^2
- ii) For a freely falling body through viscous medium, the body has ----- velocity. 1
- Critical
 - Terminal
 - Escape
 - Angular

B) Solve **any two** questions of the following.

- i) A classroom has dimension $20 \times 15 \times 5 \text{ m}^3$. The total absorption of the surface is 69 m^2 . Calculate the reverberation time. 4
- ii) Define the terms Streamline flow, Turbulent flow, Critical velocity and Ideal fluid. 4
- iii) State and explain Newton's law and Stokes Law of Viscosity. 4
